# Lenin T

Coimbatore, Tamil Nadu

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## Summary

ECE graduate with hands-on experience in software development, Python, and machine learning through academic and personal projects. Passionate about building real-world applications and exploring AI-driven solutions. Seeking entry-level opportunities in software engineering, backend development, and AI/ML.

## Education

**Kumaraguru College of Technology** **Nov 2022 – May 2026**

*Bachelor of Engineering in Electronics and Communication – CGPA: 7.90* *Coimbatore, Tamil Nadu*

**Sri Vivekananda Matriculation Higher Secondary School**

*Class XII – 86.6%* — *Class X – 96.2%* *Coimbatore, Tamil Nadu*

## Skills

**Programming & Query Languages:** Python; SQL

**Software Development:** FastAPI; REST APIs; SQLAlchemy; JWT Authentication; React; HTML; CSS; PostgreSQL; SQLite

**AI/ML:** NumPy; Pandas; Scikit-learn; Data Preprocessing; Exploratory Data Analysis; Statistics; Machine Learning; Random Forest; Model Evaluation; PyTorch; Deep Learning; Generative AI; Large Language Models (LLMs); Prompt Engineering

**Tools & Technologies:** Git; GitHub; Docker; Linux; VS Code

## Projects

**PredictIQ – Predictive Maintenance System** **Oct 2026**

*Tech Stack: Python, Scikit-learn, FastAPI, React, SQLite, Docker, Render*

- Developed a machine learning application using a Random Forest classifier to estimate machine failure probability from industrial sensor measurements.
- Built REST APIs with FastAPI and a React dashboard for predictions, probability visualization, and prediction history.
- Evaluated model performance using precision, recall, F1-score, and a confusion matrix, achieving 98.15% test accuracy on the AI4I 2020 dataset.
- Implemented SQLite-based prediction logging, API tests, Docker support, and cloud deployment on Render.

**ToLet Coimbatore – Hyperlocal Rental Platform** **Apr 2026 – Jun 2026**

*Tech Stack: Python, FastAPI, PostgreSQL, React, Cloudinary, Railway*

- Built a full-stack rental platform connecting landlords and tenants in Coimbatore through a hyperlocal listing system.
- Developed REST APIs using FastAPI and PostgreSQL for listing CRUD operations, JWT authentication, and photo uploads.
- Implemented a React frontend with map-based listing visualization using Leaflet.js and a mobile-responsive interface.
- Deployed the backend and frontend on Railway with a cloud PostgreSQL database and environment-based configuration.

**Vehicle Platooning for Heavy Duty Vehicles** **Dec 2025 – Apr 2026**

*Tech Stack: ESP32, Arduino UNO, ESP-NOW, RF Communication, Ultrasonic Sensor*

- Developed a leader-follower vehicle communication prototype using wireless communication for coordination.
- Integrated ultrasonic obstacle detection and an automatic braking mechanism.
- Implemented distance monitoring to support adaptive spacing between the leader and follower vehicles.
- Developed motor control logic using Arduino UNO and the L298N motor driver for directional movement and speed control.
- Integrated embedded controllers and wireless communication to demonstrate coordinated vehicle movement.

## Certifications

- Introduction to Artificial Intelligence (AI) – IBM, Coursera
- Introduction to Internet of Things – NPTEL (IIT Kharagpur), Elite (91%)
- Software Defined Vehicles (SDV 101) – Coursera